
AWS PrivateLink – Overview + Hands-on Lab

What is AWS PrivateLink?

AWS PrivateLink allows you to **securely access services (AWS or custom)** over the **private AWS network without using the public internet**.

 “Expose or consume services privately within or across VPCs using AWS’s internal network.”

Why Use PrivateLink?

Benefit

Description

Private access	Traffic never traverses the public internet
Security	No need for NAT, public IP, or internet gateway
Service sharing	Share internal services (like APIs, databases) privately
Cross-account	Enables secure communication across AWS accounts
Compliance	Meet strict network isolation & audit requirements

Key Components

Component	Description
Service Provider	The VPC that hosts the private service

Service Consumer	The VPC that wants to access the private service
VPC Endpoint Service	The private service exposed via PrivateLink
Interface Endpoint	ENI created in the consumer VPC to access the service

PrivateLink vs Other VPC Connections

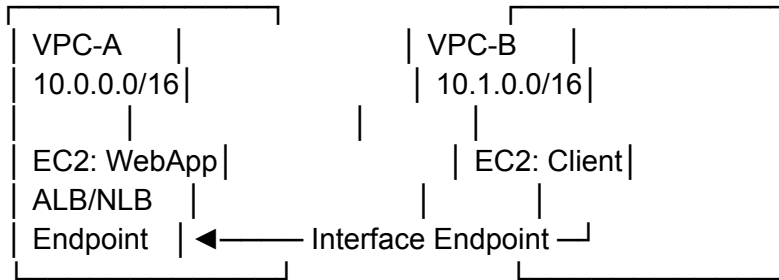
Feature	PrivateLink	VPC Peering	Transit Gateway
Traffic Isolation	✓ Yes	✗ Full routing	✗ Shared routing
Granular Access	✓ Per service port	✗ Full VPC access	✗ All routes shared
Cross-Account	✓ Native	✓ With effort	✓
Transitive Routing	✗ Not supported	✗ Not supported	✓ Supported

Lab: Use AWS PrivateLink to Access a Service in Another VPC

Goal

- Create a **custom service** (web server) in VPC-A
 - Expose it using **PrivateLink (VPC Endpoint Service)**
 - Access the service from EC2 in VPC-B via **Interface Endpoint**
-

Architecture Diagram



Step-by-Step Lab

Step 1: Create Two VPCs

VPC	CIDR Block	Purpose
VPC-A	10.0.0.0/16	Service Provider
VPC-B	10.1.0.0/16	Service Consumer

Step 2: Create Service in VPC-A

1. Launch EC2 instance (EC2-A) in VPC-A

Install NGINX:

```
sudo yum install nginx -y
sudo systemctl start nginx
```

- 2.
3. Confirm it listens on port 80
4. (Optional) Add an NLB in front of EC2-A

✓ Step 3: Create Endpoint Service (in VPC-A)

1. Go to **VPC** → **Endpoint Services** → **Create**
 2. Name: `MyPrivateService`
 3. Choose the **NLB** (or EC2 ENI)
 4. Enable:
 - "Require acceptance"
 - Cross-account (if needed)
-

✓ Step 4: Create Interface Endpoint (in VPC-B)

1. Go to **VPC** → **Endpoints** → **Create Endpoint**
 2. Service Name: Find `com.amazonaws.vpce.<region>.vpce-svc-xxxx`
 3. Type: **Interface**
 4. Subnet: Select private subnet in VPC-B
 5. Security group: allow TCP 80
 6. Accept the connection in the provider account (VPC-A)
-

✓ Step 5: Test from EC2 in VPC-B

1. Launch EC2 in VPC-B
2. SSH into EC2

Use curl:

```
curl http://<PrivateLink-DNS-name>
```

3.

✓ You'll see the NGINX welcome page or your custom service response.

Cleanup

- Delete interface endpoints
 - Delete endpoint service
 - Terminate EC2s
 - Delete VPCs and networking
-

✓ Summary

Component	Purpose
VPC Endpoint Service	Exposes your service privately
Interface Endpoint	Connects consumer to provider
NLB or ENI	Backend of the private service
Private DNS	Automatically maps endpoint to service

Real-World Use Cases

- Internal API access between teams/accounts
 - Exposing SaaS services securely (e.g., Datadog, Snowflake)
 - Replace public API Gateway endpoints with private access
-

